Set_A_Lab1 - Results

Attempt 1 of 1

Written Jan 14, 2026 12:11 PM - Jan 14, 2026 4:53 PM

Attempt Score	**12 / 15 - 80 %**
Overall Grade (Highest Attempt)	**12 / 15 - 80 %**

Question 1

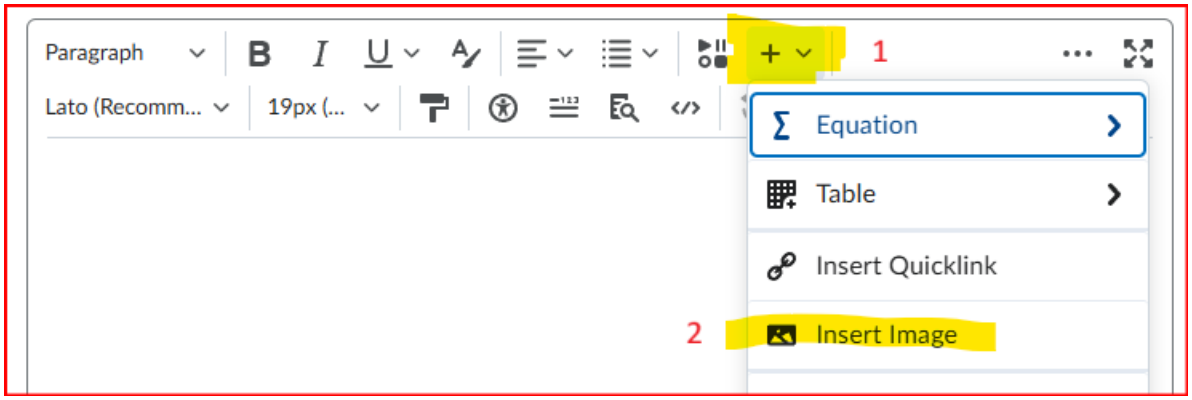
Consider the following C code (*main.c*) and answer/complete the following questions/tasks (Q1, Q2, Q3, Q4, Q5, Q6, Q7):

```c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf(" Preocess ID : %d \n", getpid());
    sleep(1);
    return 0;
}
```

Use the online C compiler https://www.onlinegdb.com/online_c_compiler and run your code. Insert a screenshot of your output.

Hint:


Paragraph
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main.c

```c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf(" Process ID : %d\n", getpid());
    sleep(1);
    return 0;
}
```

```
Process ID : 7747


...Program finished with exit code 0
Press ENTER to exit console.
```

1.png (13.93 KB)

The correct answer is not displayed for Written Response type questions.

Question 2

Use your Linux environment https://cocalc.com/features/linux and run your code. Insert a screenshot of your output.

Hint:

gcc main.c -o main.out

./main.out

The correct answer is not displayed for Written Response type questions.

Question 3

In your code change the sleeping time from **1s** to **30s**, recompile your code and in the Linux terminal run the following command:

./main.out & pstree –p -T

When you run *./main.out & and patree -p -T*, do both outputs display the same PID for *main.out* (Y/N)? ___N___ ✖ **(Y)**

Question 4

After running `pstree –p –T`, what does the `–p` option show?

- ✔ (●) Process IDs (PIDs)
- ( ) Process names
- ( ) Parent directories
- ( ) Ports

Question 5

After changing the sleep duration from `1s` to `30s`, running:

`./main.out &`

does what?

- ○ Kills all other background processes
- ✔ ◉ Runs the program in the background and returns control to the terminal
- ○ Runs the program in debug mode
- ○ Runs the program with root permissions

Question 6

After running *pstree -p -T*, what does the *-T* option do?

- ○ Set a tree depth limit
- ○ Show timestamps
- ➡ ○ Hide threads and show only processes
- ✖ ◉ Show thread information

Question 7

When running ./main.out in two different terminals, which of the following is true?

- ○ The PID is always 1
- ✔ ◉ Each run creates a new, unique PID
- ○ The PID depends on CPU mode
- ○ Both terminals will show the same PID

Question 8

Use your Linux environment https://cocalc.com/features/linux and run *lscpu* command and answer/complete the following questions (Q8, Q9, Q10, Q11, Q12).

Insert a screenshot of the entire *lscpu* output.

```
  Address sizes:         46 bits physical, 48 bits virtual
  Byte Order:            Little Endian
CPU(s):                  4
  On-line CPU(s) list:   0-3
Vendor ID:               GenuineIntel
  Model name:            Intel(R) Xeon(R) CPU @ 2.80GHz
    CPU family:          6
    Model:               85
    Thread(s) per core:  2
    Core(s) per socket:  2
    Socket(s):           1
    Stepping:            7
    BogoMIPS:            5599.99
    Flags:               fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xt
opology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch invpcid_single s
sbd ibrs ibpb stibp ibrs_enhanced fs
                         gsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm avx512f avx512dq rdseed adx smap clflushopt clwb avx512cd avx512bw avx512vl xsaveopt xsavec xgetbv1 xsa
ves arat avx512_vnni md_clear arch_capabilities
Virtualization features:
  Hypervisor vendor:     KVM
  Virtualization type:   full
Caches (sum of all):
  L1d:                   64 KiB (2 instances)
  L1i:                   64 KiB (2 instances)
  L2:                    2 MiB (2 instances)
  L3:                    33 MiB (1 instance)
NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-3
Vulnerabilities:
  Gather data sampling:  Not affected
  Itlb multihit:         Not affected
  L1tf:                  Not affected
  Mds:                   Not affected
  Meltdown:              Not affected
  Mmio stale data:       Vulnerable: Clear CPU buffers attempted, no microcode; SMT Host state unknown
  Reg file data sampling: Not affected
  Retbleed:              Mitigation; Enhanced IBRS
  Spec rstack overflow:  Not affected
  Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl and seccomp
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; RSB filling; PBRSB-eIBRS SW sequence; BHI SW loop, KVM SW loop
  Srbds:                 Not affected
  Tsx async abort:       Vulnerable: Clear CPU buffers attempted, no microcode; SMT Host state unknown
~/A$
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```

1.png (118.75 KB)

The correct answer is not displayed for Written Response type questions.

Question 9

Use your *lscpu* output and complete the following.

✔ __1__ Architecture

1. x86_64

✔ __8__ Byte Order

✔ __3__ CPU op-mode(s)

✔ __6__ Address sizes

2. x86_32

3. 32-bit, 64-bit

4. 32-bit

5. 64-bit

6. 46 bits physical, 48 bits virtual

7. 48 bits physical, 46 bits virtual

8. Little Endian

9. Big Endian

Question 10

Run `lscpu | grep cache` command in your terminal and insert a screenshot of your output.

```
~/A$ lscpu | grep cache
L1d cache:                           64 KiB (2 instances)
L1i cache:                           64 KiB (2 instances)
L2 cache:                            2 MiB (2 instances)
L3 cache:                            33 MiB (1 instance)
~/A$
```

1.png (6.46 KB)

The correct answer is not displayed for Written Response type questions.

Question 11

When running the command: `lscpu | grep cache` , what is the purpose of grep cache?

- It sorts the CPU information alphabetically
- ✔ It filters the output to show only lines containing the word "cache"
- It deletes all cache-related information
- It displays cache usage in real time

Question 12

In the command: lscpu | grep cache what role does the pipe symbol (|) play?

- It redirects errors to a log file
- It pauses lscpu before executing grep
- ✔ It combines two commands so the output of lscpu becomes the input of grep cache
- It runs both commands at the same time

Done